Watchdog Trust for Byzantine-Robust Federated Learning in VANETs: A Measurable Breakdown Condition, and Where It Fails

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***Abstract*—Byzantine-robust aggregation protects federated learning (FL) only while attackers are a minority of the updates in a round. In vehicular networks attackers are rare on the road, but their share among the updates of one round can still be large: rounds are local, small and biased towards vehicles that stay. Vehicles that poison a shared detector often also misbehave on the network, where neighbours can observe them. We study a filter that scores each vehicle by watchdog observations of its packet forwarding and discards low-trust updates before robust aggregation. The filter turns the breakdown point of the aggregator into a function of two measurable rates, the attacker pass rate β and the honest rejection rate γ: robust aggregation holds while $\beta/(1-\gamma) < (1-\rho)/\rho$. In ns-3 simulations with grayholes that really drop transit packets, non-IID clients and vehicle-disjoint evaluation, the filter raises no false alarms and costs no accuracy when attackers are rare (0–10%), and under sign-flip, label-flip and ALIE poisoning it keeps every trust-filtered rule at the no-attack F1 up to a 70% attacker share, while median, trimmed mean and Krum each fail against at least one attack from 40–60%. The condition predicts whether trust-filtered median survives in all 56 settings we test. It also maps the limits: weak and on-off grayholes raise β; slow trust recovery with a 30 s observation period repairs the on-off case, and a per-round analysis explains why the observation period is needed. Poisoners that forward honestly defeat the filter.**

***Index Terms*—Federated learning, VANET, grayhole attack, Byzantine robustness, trust management, watchdog, ns-3.**

I. Introduction

Grayhole attacks, in which a vehicle that should relay packets silently drops some of them, remain a practical threat to multi-hop vehicular ad hoc networks (VANETs) [1]. Learning-based detectors trained on network features can identify such vehicles, and federated learning (FL) [2] lets vehicles train a shared detector without uploading their observations. FL is itself exposed to poisoning by the same malicious vehicles [3], [4]. Byzantine-robust rules such as Krum [5], coordinate-wise median and trimmed mean [6] tolerate poisoned updates only while attackers are a minority of the updates they aggregate, and their non-IID refinements [7] need a margin below that.

Attackers are rare on a real road, so why worry about majorities? Because the share that matters is the share *in a round*, not on the road. Vehicular FL rounds are local (the vehicles near one roadside unit or in one area), often small, and biased towards vehicles that stay in coverage, which a parked attacker does and a passing honest vehicle does not. A few colluding attackers can therefore be a majority of one round while being a few percent of all traffic, and random client sampling alone produces such rounds [8].

Vehicular FL also has a resource that generic FL lacks: the network is observable. A vehicle that drops packets it should forward can be caught by its neighbours with a watchdog [9]. Vehicular FL schemes already weight or select clients by reputation [10]–[14], but they compute it from the model updates or from interaction history, which shares the blind spots of the aggregator. We ask: *if trust comes from observable network behaviour, when does it keep Byzantine-robust FL working, at what cost when attackers are rare, and where does it stop?* We answer with a pipeline built to avoid shortcuts: attackers really drop transit packets in ns-3, features come from what a watchdog can actually overhear (including false accusations from collisions and mobility), the threshold is label-free, trust uses only past observations, clients are non-IID, and evaluation is vehicle-disjoint 3-fold cross-validation.

Contributions.

1) *A breakdown condition in observable quantities.* The filter maps the attacker share ρ to an aggregated share ρ_{surv} that depends only on its pass and rejection rates (Proposition 1); the breakdown point becomes $(1-\gamma)/(1-\gamma+\beta)$, 0.95 for the filter we measure. $\rho_{\mathrm{surv}} < 1/2$ predicts the survival of trust-filtered median in all 56 settings (Fig. 1).
2) *Robustness that does not depend on the poisoning attack, at no accuracy cost when attackers are rare.* With 0–10% attackers the filter matches unfiltered training and raises no false alarms; up to a 70% share it keeps all trust-filtered rules at the no-attack F1 under sign-flip, label-flip and ALIE [15], whereas each unfiltered robust rule fails against at least one of them.
3) *A map of failure modes and two fixes.* Weak and on-off grayholes raise β; slow trust recovery with a short observation period restores the on-off case, and a per-round analysis of 1,260 runs shows that failures the pooled condition misses happen before attackers are detected. Honest-forwarding poisoners defeat the filter, which then amplifies them.

TABLE I
TRUST IN VEHICULAR FL: WHERE THE SCORE COMES FROM AND WHAT IS ANALYSED.

Work	Trust signal	Indep.[a]	Evaluation	Breakdown[b]
[10]	Reputation from FL contributions	–	IoV sim.	–
[11]	Communication behaviour, model transmissions	partly	IoV sim.	–
[13]	Reputation, context, resources	partly	MNIST	–
[12]	Accuracy, contribution frequency, anomalies	–	HFL sim.	–
[14]	Update consistency, anomaly deviation	–	VeReMi	–
Ours	Watchdog-observed packet forwarding	✓	ns-3, real drops	✓

[a] Signal independent of the model updates. [b] Quantifies how filter errors (β, γ) reach the aggregator and where it breaks.

II. RELATED WORK

Byzantine-robust FL. Krum [5] selects the update closest to its neighbours; median and trimmed mean [6] act per coordinate; bucketing [7] averages random groups first to reduce heterogeneity. All assume an honest majority among aggregated updates; ALIE [15] evades them even then, and client subsampling can create Byzantine-majority rounds [8]. Defences that tolerate an attacker majority need trusted data at the server: Zeno [16] scores updates on a validation set and FLTrust [17] on a clean root dataset, while FLAME [18] clusters and clips updates under an honest majority. Our filter needs no server data; its signal is the network.

Reputation and trust in vehicular FL. Trust management in VANETs is surveyed in [19], and evicting misbehaving vehicles in [20]. Reputation-based selection with blockchain [10], trust-aware client selection with mobility and channel context [11], [13], anomaly records in hierarchical FL [12] and update-consistency trust [14] compute trust from model updates, contribution history or link reliability, usually assume an attacker minority (at most 45% in [10]), and do not quantify how the filter's errors reach the aggregator (Table I). Our trust comes from an independent channel, network forwarding, and we give the exact propagation.

Watchdogs. Overhearing whether a relay forwards a packet is the classic defence against selective forwarding [9]; its false positives and negatives under collisions and mobility [21] are what make $\gamma > 0$ here, and on-off attacks against trust are known [22], [23]. We measure both inside FL.

III. SYSTEM AND THREAT MODEL

N vehicles form a multi-hop VANET with AODV routing [24]. Each runs a watchdog and periodically reports its observations; a server (e.g., at the edge) turns the reports into per-vehicle trust and coordinates FL rounds in which vehicles train the shared detector and upload their models. We write ρ for the attacker share among the vehicles of an FL round. On the road ρ is small; locally or in a round it can be large (Section I). We do not model how rounds become attacker-heavy; we evaluate the filter over round-level shares from 0 to 0.7.

Attackers are *aligned*: each acts as a grayhole, dropping transit data packets with probability p_d while taking part normally in route discovery, and poisons its FL updates. We consider one data-poisoning and two model-poisoning attacks: under *label-flip* [4] each attacker trains on its data with every label inverted; under *sign-flip* each sends $w_g - \lambda(w - w_g)$, where w_g is the global model, w its locally trained model and $\lambda = 5$; under non-omniscient *ALIE* [15] all send $\mu - z\sigma$, the per-coordinate mean minus z standard deviations of the attackers' own honest updates. Harder network behaviours are *weak* grayholes ($p_d \in \{0.2, 0.3\}$), *on-off* grayholes (drop 90% for 10 s, forward honestly for 20 s, random phase), and *decoupled* poisoners drawn from vehicles that forward honestly.

Out of scope. Attackers hold one identity each (no Sybil identities), do not forge route replies (a blackhole that attracts routes is easier to detect), and do not tune their dropping to the threshold; denial of service and inference attacks are not considered. Trust-aware attackers, Sybil identities and authenticated uploads are studied in follow-up work.

IV. TRUST-FILTERED AGGREGATION

Watchdog. When vehicle A hands a data frame to relay X (network destination not X), it opens an entry; the entry counts as forwarded if A decodes X re-transmitting the same packet within 1 s. A watchdog cannot tell a drop from a missed overhearing, so honest vehicles score below one. Per vehicle and second we export cumulative counters (relay requests, overheard forwards, route requests received and rebroadcast, packets and bytes, delay, neighbours); ground truth goes only to a separate summary.

Windowed trust. At round time t the trust of vehicle v is $T_v = 0.7 f_v + 0.3 r_v$, with f_v its overheard forwarding ratio and r_v its route participation (route requests rebroadcast over received), both from counter differences over $(t - W, t]$, $W = 10$ s. A ratio with fewer than five events keeps its last value; never-observed vehicles are accepted. As a slow-recovery variant, *trust memory* M takes the minimum windowed trust over $(t - M, t]$, so a vehicle caught dropping stays suspect for M seconds (slowed recovery after misbehaviour is also used in [11]; we measure its effect on β and on the aggregator); with a *warm-up* τ_w, updates are aggregated only after trust has observed the network for τ_w seconds.

Label-free threshold. Each round a two-component Gaussian mixture is fitted to the trust values and the low component is rejected; if the Bayesian information criterion [25] prefers one component, all vehicles are accepted.

Aggregation. Kept updates are combined by trust- and size-weighted averaging (Trust-FLAD, TF), Krum (TF+Krum, $f = \lfloor n/4 \rfloor$), coordinate-wise median (TF+Median) or bucketed variants (bucket size 2).

V. When Does the Filter Restore Robustness?

Let β be the probability that the filter keeps an attacker and γ the probability that it rejects an honest vehicle. The attacker share among aggregated updates is

$$\rho_{\mathrm{surv}} = \frac{\rho\beta}{\rho\beta + (1-\rho)(1-\gamma)}. \tag{1}$$

Proposition 1. *$\rho_{\mathrm{surv}} < 1/2$ if and only if $\beta/(1-\gamma) < (1-\rho)/\rho$. Hence a rule that is robust while attackers are a minority of its inputs stays robust after the filter for every $\rho < \rho^* = (1-\gamma)/(1-\gamma+\beta)$.*

Proof. $\rho_{\mathrm{surv}} < 1/2 \iff \rho\beta < (1-\rho)(1-\gamma)$; solve for ρ. □

The left side depends only on how well network behaviour separates attackers from honest vehicles, the right side only on the attacker share. With $\beta = 0.05$ and $\gamma = 0.10$, $\rho^* = 0.95$; with $\beta = 0.5$ (on-off attackers) $\rho^* = 0.65$. Three remarks qualify the condition. (i) β rises with ρ when separation is weak, because the honest reference population shrinks. (ii) For rules that need a margin (Krum, bucketing) $\rho_{\mathrm{surv}} < 1/2$ is necessary, not sufficient; Krum can also survive $\rho_{\mathrm{surv}} > 1/2$ when poisoned updates are scattered outliers. (iii) β and γ are run averages: an attacker passes until the watchdog has seen it misbehave, so the condition describes the filter after a detection transient (Section VII-F).

VI. Simulation Setup

Network. ns-3.42 [26]; 50 vehicles on a 2 km two-lane ring road at 20–30 m/s (constant density, no partitions); IEEE 802.11a at 6 Mb/s, unit-disk propagation at 250 m, AODV; 30 UDP flows (512 B, 50 kb/s) between random pairs; 100 s per run. Attackers are a seeded random subset and drop transit data in a routing layer above AODV, leaving the control plane untouched. Packet delivery is 0.90 without attackers and 0.44–0.82 with $p_d = 0.5$; the watchdog forwarding ratio averages 0.93 for honest vehicles and 0.45 for grayholes.

Learning. The detector is an MLP (16 features, 32–16–1) that classifies a vehicle-second as attacker or not. A client's data are the rows of the vehicles in its radio range (observed, non-IID clients). Vehicles are split into three stratified folds whose rows are held out from all clients; predictions are pooled so every vehicle is tested once. FL runs 20 rounds of 3 local epochs (Adam, learning rate 0.01, batch 128). Attacker shares $\rho \in \{0, 0.04, 0.1, 0.2, \ldots, 0.7\}$ (0 to 35 of 50 vehicles), five seeds each. We report F1 of the attacker class against the ceiling, FedAvg without poisoning on the same network. Code and data: https://github.com/MAZHAR1807102/watchdog-trust-fl-vanet.

VII. Results

A. Rare attackers: no accuracy cost

With no attackers the filter raises no false alarms: every rule, filtered or not, classifies all vehicles as honest (accuracy 1.00). It is not free, though: the mixture rarely concludes that there is no second population, so it still rejects 12% of the honest vehicles per round (44 of 50 kept), a participation cost but not an accuracy cost. With 5 attackers of 50 (10%), poisoning does no visible damage to any rule and every rule, with or without the filter, stays at the ceiling (0.59–0.71 against 0.66, Table II); the filter passes 4% of the attackers and rejects 11% of the honest vehicles. With only 2 attackers (4%) the task itself cannot be learned (no-attack F1 0.15), so we omit that share from the table. At 20–30% median and trimmed mean also match the ceiling. In short, when attackers are rare robust aggregation alone suffices, and the filter neither helps nor hurts accuracy; its value lies in the rounds where attackers are many.

B. Attacker majorities: three poisoning attacks

From 40–50% on the filter's value appears (Table II). It passes 5–6% of the attackers and rejects 8–12% of the honest vehicles at every share, so even at 70% only about 2 attackers are aggregated against about 14 honest vehicles ($\rho_{\mathrm{surv}} \approx 0.13$), and from 20% on all five trust-filtered rules stay within 0.01–0.03 of the ceiling under both attacks. Without the filter every rule fails against some attack. Under sign-flip, trimmed mean breaks at 40% and median at 50%, while Krum survives to 60%: $\times 5$ sign-flipped updates are scattered outliers it never selects (poisoned picks 0% up to 60%, 31% at 70%). Label-flip is harmless below 40% but breaks every unfiltered rule at 50–60%, Krum included (0.03 at 60–70%): label-flipped models are mutually consistent, so an attacker majority forms the tightest cluster and Krum picks it. Bucketing hurts without the filter, as more buckets contain an attacker (37% at $\rho = 0.2$, 92% at 0.7). ALIE, built to fool distance- and coordinate-based rules, does so for $z \geq 2$: at 50/60/70% Krum and median fall to 0.09/0.32/0.29 with $z = 3$ and FedAvg to 0.49 at 70%, while the trust-filtered rules stay at 0.79–0.97, the ceiling, because the filter removes the attackers before the aggregator sees ALIE's tight cluster. After filtering, robustness does not depend on the poisoning attack; without the filter, which rule survives does.

C. The breakdown condition predicts survival

Fig. 1 plots, for every setting with a learnable task, the relative F1 of TF+Median against ρ_{surv} computed from the measured β and γ. All 56 points fall on the predicted side of $\rho_{\mathrm{surv}} = 1/2$: the 48 settings with $\rho_{\mathrm{surv}} < 0.5$ keep at least 70% of the no-attack F1 (43 of them at least 97%), and all 8 with $\rho_{\mathrm{surv}} > 0.5$ collapse to 0–36%. The breakdown of a trust-filtered robust rule is predictable from two rates that can be measured on the network.

D. Ablation

Table III varies one choice at a time. Otsu's threshold rejects as few attackers as the mixture but cuts inside the honest mode: it rejects 55–76% of honest vehicles, leaves about nine clients per round, and TF+Median fails in one of five seeds at 70%. Without rejection, TF+Median is plain median and collapses from 50%, and the trust-weighted mean alone fails from 40%, so the gain comes from rejection, not weighting. Windows of 5–20 s give the same result, and the 10 s windowed filter already matches trust computed from the whole run.

TABLE II

F1 BY ATTACKER SHARE, $p_d = 0.5$ (MEAN OF 5 SEEDS, POOLED 3-FOLD; 40% AND 60% OMITTED FOR SPACE, QUOTED IN THE TEXT). TF = TRUST-FILTERED; β, γ OF THE FILTER.

	Sign-flip ($\lambda = 5$)					Label-flip				
Rule	10%	20%	30%	50%	70%	10%	20%	30%	50%	70%
No-attack ceiling	0.66	0.80	0.87	0.92	0.96	0.66	0.80	0.87	0.92	0.96
FedAvg	0.68	0.33	0.41	0.06	0.24	0.71	0.81	0.82	0.55	0.10
Krum	0.59	0.72	0.87	0.78	0.69	0.60	0.73	0.86	0.46	0.03
Trimmed mean	0.67	0.79	0.84	0.08	0.00	0.69	0.80	0.86	0.45	0.06
Median	0.71	0.79	0.86	0.39	0.00	0.71	0.81	0.86	0.50	0.04
Krum+Bucket	0.65	0.78	0.84	0.42	0.09	0.66	0.81	0.87	0.62	0.03
Median+Bucket	0.66	0.80	0.47	0.02	0.19	0.70	0.81	0.82	0.55	0.10
TF	0.67	0.79	0.86	0.92	0.97	0.67	0.80	0.86	0.91	0.97
TF+Krum	0.61	0.79	0.85	0.90	0.94	0.61	0.79	0.85	0.90	0.95
TF+Median	0.70	0.80	0.87	0.91	0.97	0.70	0.79	0.86	0.91	0.96
TF+Krum+Bucket	0.65	0.80	0.87	0.91	0.97	0.66	0.80	0.87	0.91	0.96
TF+Median+Bucket	0.67	0.79	0.86	0.92	0.97	0.67	0.80	0.86	0.91	0.96
β	0.04	0.05	0.05	0.05	0.06	0.04	0.05	0.05	0.05	0.06
γ	0.11	0.10	0.12	0.10	0.08	0.11	0.10	0.12	0.10	0.08

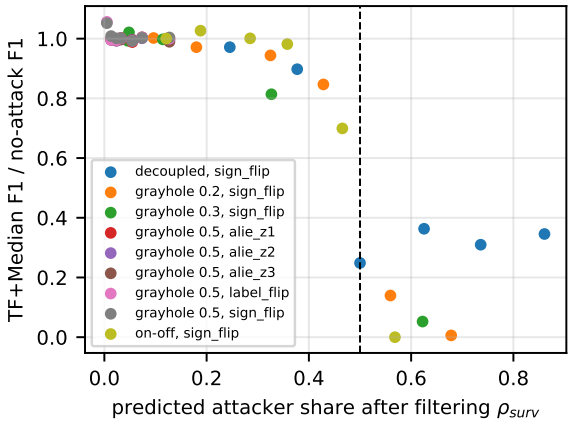


Fig. 1. Relative F1 of TF+Median (F1 over the no-attack F1 of the same network) versus ρ_{surv} predicted by (1) from measured β, γ and ρ, for 56 settings (attacker type, poisoning, ρ; settings whose no-attack F1 is below 0.5 are omitted).

TABLE III

FILTER ABLATION, SIGN-FLIP, $p_d = 0.5$: TF+MEDIAN F1 AND FILTER RATES OVER $\rho = 20$–70%. STATIC TRUST USES THE WHOLE RUN, INCLUDING THE FUTURE (AN UPPER BOUND).

Variant	50%	60%	70%	β	γ
Mixture, $W = 10$ s (ours)	0.91	0.97	0.97	0.05–0.06	0.08–0.12
Otsu threshold [27]	0.91	0.96	0.77	0.03–0.04	0.55–0.76
No filter (= median)	0.39	0.01	0.00	1	0
$W = 5$ s / $W = 20$ s	0.91	0.96	0.97	0.04–0.07	0.06–0.11
Static trust (oracle)	0.91	0.97	0.96	0.00	0.00–0.08

E. Failure modes

The filter can only reject what the watchdog separates (Fig. 2). **Weak grayholes** ($p_d = 0.2$–0.3) overlap honest vehicles that lose overhearings to collisions, so β grows with ρ (Table IV) and TF+Median collapses where ρ_{surv} crosses one half (60–70%); TF+Krum degrades gracefully (0.69–0.91) because the sign-flipped updates that pass remain outliers. **On-off grayholes** keep a high windowed trust half of the time ($\beta \approx 0.5$): TF and TF+Median collapse once $\rho_{\text{surv}} > 1/2$, TF+Krum holds. A trust memory of 60 s keeps an attacker suspect through its honest phase and cuts β to 0.12–0.20 (at a γ of 0.10–0.19); TF+Median then holds to 60%, but at 70% it still fails in three of five seeds although the pooled $\rho_{\text{surv}} \approx 0.34$. A 30 s warm-up removes these failures: β falls to 0.07–0.09 and TF+Median reaches 0.77/0.84/0.88 at 50/60/70%, all five seeds at 70% between 0.84 and 0.90

TABLE IV

HARDER ATTACKERS, SIGN-FLIP (F1, MEAN OF 5 SEEDS; ρ = 40 / 50 / 60 / 70%).

Attacker	Quantity	40 / 50 / 60 / 70%
grayhole, $p_d = 0.3$	ceiling	0.86 / 0.92 / 0.94 / 0.96
	β	0.18 / 0.11 / 0.29 / 0.67
	TF+Median	0.86 / 0.92 / 0.77 / 0.05
	TF+Krum	0.81 / 0.91 / 0.91 / 0.82
grayhole, $p_d = 0.2$	ceiling	0.77 / 0.86 / 0.90 / 0.91
	β	0.66 / 0.70 / 0.80 / 0.86
	TF+Median	0.73 / 0.73 / 0.13 / 0.01
	TF+Krum	0.72 / 0.79 / 0.82 / 0.69
on-off (10 s / 20 s)	ceiling	0.71 / 0.78 / 0.85 / 0.88
	β	0.54 / 0.51 / 0.53 / 0.54
	TF+Median	0.71 / 0.76 / 0.59 / 0.00
	TF+Krum	0.72 / 0.77 / 0.77 / 0.79
on-off, memory 60 s	β	0.18 / 0.17 / 0.18 / 0.20
	TF+Median	0.71 / 0.77 / 0.84 / 0.27
	TF+Krum	0.72 / 0.76 / 0.80 / 0.84
on-off, memory 60 s + warm-up 30 s	β	– / 0.08 / 0.07 / 0.09
	TF+Median	– / 0.77 / 0.84 / 0.88
	TF+Krum	– / 0.77 / 0.80 / 0.86
decoupled poisoners	β	0.91 / 0.91 / 0.90 / 0.90
	γ	0.39 / 0.45 / 0.52 / 0.66
	best rule	0.73 / 0.43 / 0.34 / 0.33

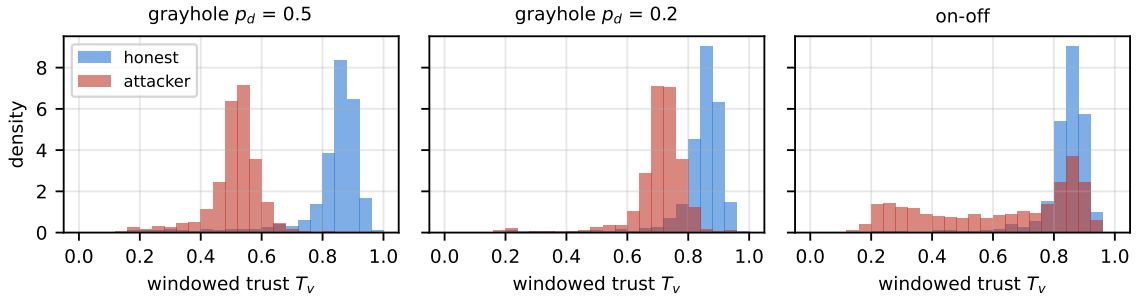


Fig. 2. Windowed trust of honest vehicles and attackers ($\rho = 0.3$, 5 seeds, every 5 s). Strong grayholes form a separate mode; weak grayholes overlap the honest mode; on-off attackers spread across the range.

(ceiling 0.88). A 30 s memory is weaker than 60 s and 90 s is no better, a 60 s warm-up adds nothing over 30 s, and trust over the whole past dilutes bursts ($\beta = 0.16$ at 70% even with warm-up). **Decoupled poisoners**, which forward honestly, pass the filter ($\beta \approx 0.9$) while the honest-training grayholes are rejected, so the poisoner share among kept updates *rises*; all rules fail from 40%. A network-layer filter helps only when network and model misbehaviour come from the same vehicles.

F. The detection transient

Why does the pooled condition miss the on-off failures? For 1,260 TF+Median runs (setting, seed, fold) of the sign-flip and ALIE settings, the ablation and the harder attackers, we computed the attacker share among accepted updates in every round. The pooled test $\rho_{\text{surv}} \geq 1/2$ classifies 96.3% of the runs correctly (collapse: final F1 below half the ceiling) but misses 45 of 101 collapses. These share one pattern: attackers are the majority of the accepted updates in the first rounds, before the watchdog has caught them, and median aggregation drives the model into a state that the remaining honest-majority rounds do not repair. With on-off attackers at 70% and a 60 s memory, attackers hold 62–70% of round 1 and a majority in about four of the first five rounds; one seed recovers at round 15, the others never. Persistent grayholes are caught within one or two rounds and the model recovers at once (F1 0.95 from round 3). Adding "attacker majority in at least four of the first five rounds" to the pooled test raises the accuracy to 98.3% (18 misses, 4 false alarms). Rejecting never-observed vehicles does not help, because on-off attackers have been observed, in their honest phase. Robustness thus needs $\rho_{\text{surv}} < 1/2$ after the detection transient, and an observation period longer than the watchdog needs to see a burst.

VIII. Discussion and Limitations

Why FL if trust flags attackers? The filter needs neighbour reports aggregated over a window and acts only on observed vehicles; the learned detector runs on each vehicle's own per-second features and generalises to vehicles not yet scored. Both use network features, so applying the filter to an FL task with different inputs (e.g., misbehaviour detection from position messages) is the natural next step.

Overhead. Fitting the mixture takes 2–5 ms per round for 50 to 5,000 vehicles, the memory is a minimum over about a dozen stored windows per vehicle, and the watchdog is passive overhearing with a table lookup. The real costs are participation (updates lost to γ, 10–12% even without attackers, and to the warm-up) and the reports, about a dozen counters per vehicle per second.

Limitations. The radio is a unit disk and the road a ring rather than a SUMO trace, at 20–30 m/s in one direction; higher relative speeds shorten contacts and will raise γ. Every vehicle joins every round, so the round-level attacker share is set directly rather than produced by coverage and uploads. Five seeds per setting give 95% intervals of about ± 0.01–0.03 for the trust-filtered rules; collapsing rules are bimodal across seeds. Attackers do not forge route replies. The watchdog assumes promiscuous overhearing. An on-off attacker whose honest phase exceeds the memory regains trust, at the price of dropping less. Final-model F1 hides the transient: models taken during the first rounds can be poisoned.

IX. Conclusion

Network-layer trust keeps Byzantine-robust FL working when attackers become a local or temporary majority of a round, and costs no accuracy when they are rare. With watchdog trust, a label-free mixture threshold and a sliding window, trust-filtered rules stay at the no-attack F1 up to a 70% attacker share under sign-flip, label-flip and ALIE poisoning, while every unfiltered rule fails against at least one of them. The breakdown point becomes $(1-\gamma)/(1-\gamma+\beta)$, a function of two rates measurable on the network, and $\rho_{\text{surv}} < 1/2$ predicted survival in all 56 settings we tested. The defence ends where the watchdog cannot see: weak grayholes erode the separation, on-off grayholes need slow trust recovery and an observation period, and honest-forwarding poisoners evade the filter and are amplified by it.

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